Hellenic Complex Systems Laboratory

Analysis of Diagnostic Accuracy Measures for Two Combined Diagnostic Tests

Technical Report XI

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Short Description of the Demonstration

This Demonstration shows plots of diagnostic accuracy measures for two combined diagnostic tests applied at a single point in time to nondiseased and diseased populations. The calculations are performed for different values of disease prevalence, population means and standard deviations, and the respective correlation coefficients. The means and standard deviations are expressed in arbitrary units. The user can select the following measures of the combined tests using the "plot" popup menu:

• CSe: sensitivity

• CSp: specificity

• PPV: positive predictive value

• NPV: negative predictive value

• OR: (diagnostic) odds ratio

• $LR+$: likelihood ratio for a positive test result

• $LR-$: likelihood ratio for a negative test result

These measures are plotted against the sensitivities or specificities of each single test, selected using the corresponding "versus" button.

Search Terms

sensitivity, specificity, diagnostic test, diagnostic accuracy, normal distribution, bivariate normal distribution, positive predictive value, negative predictive value, likelihood ratio, odds ratio

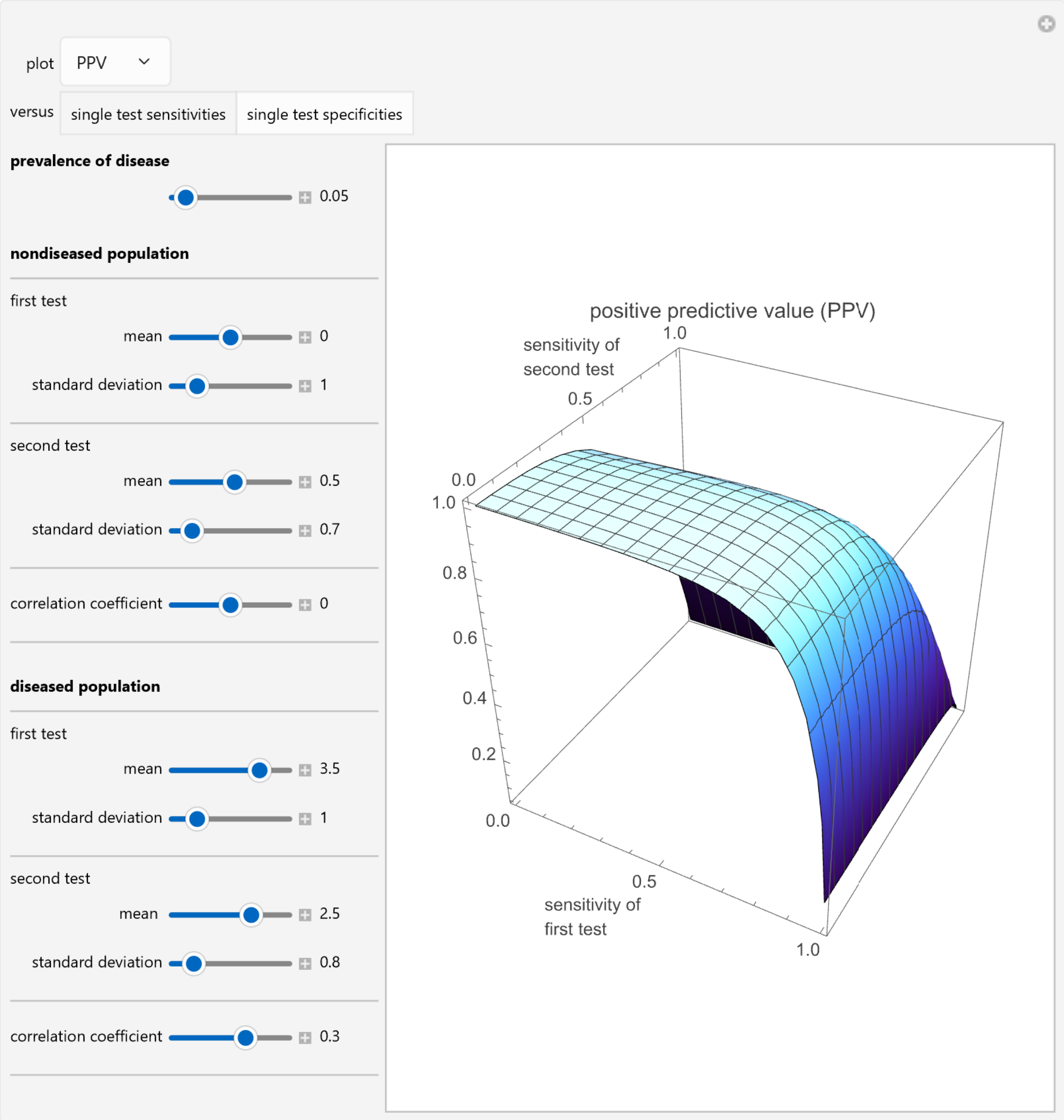

Figure 1: *Plot of the positive predictive value of two combined diagnostic tests plotted against the sensitivity of each single test, with the settings shown on the left..*

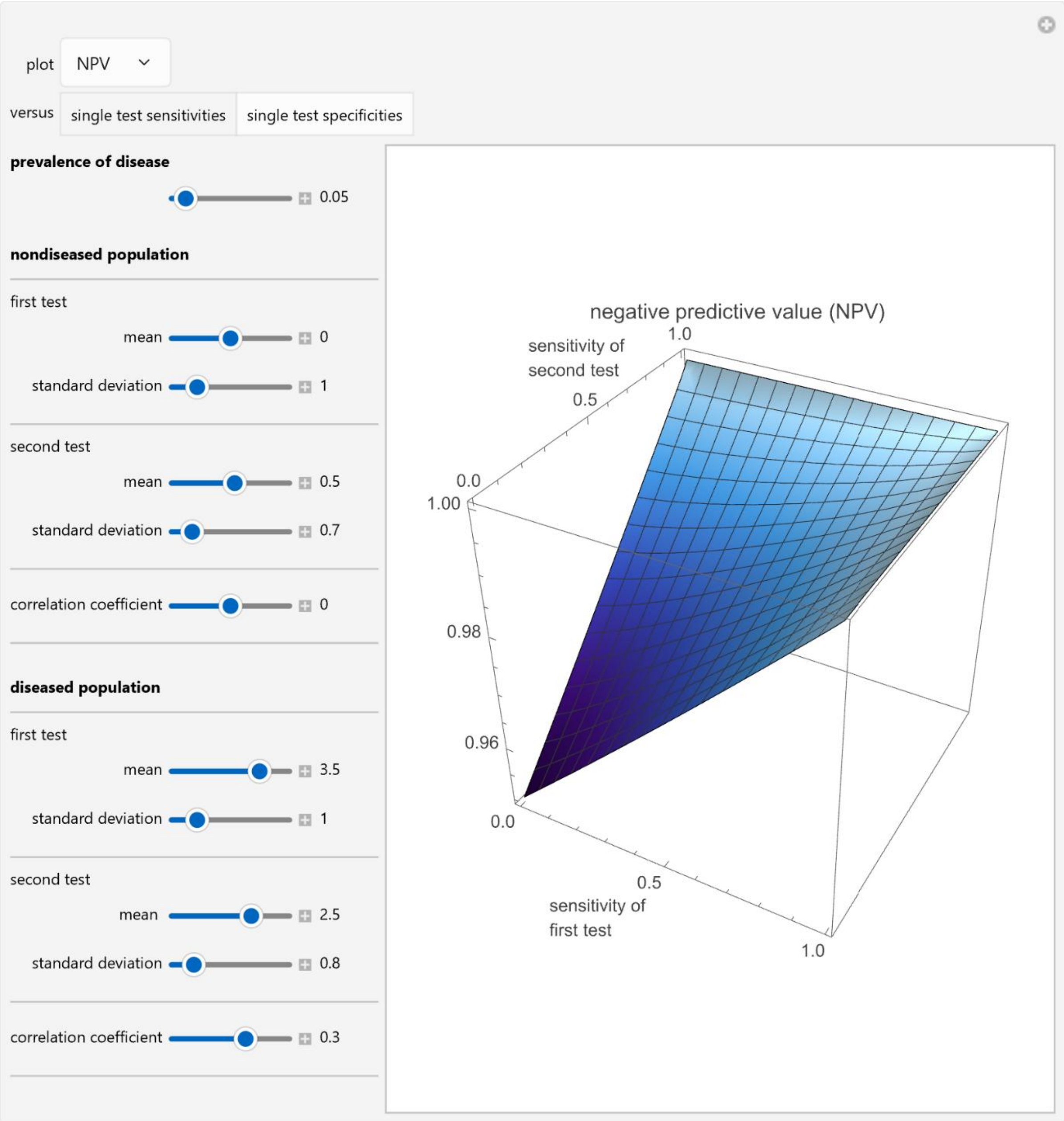

Figure 2: *Plot of the negative predictive value of two combined diagnostic tests plotted against the sensitivity of each single test, with the settings shown on the left.*

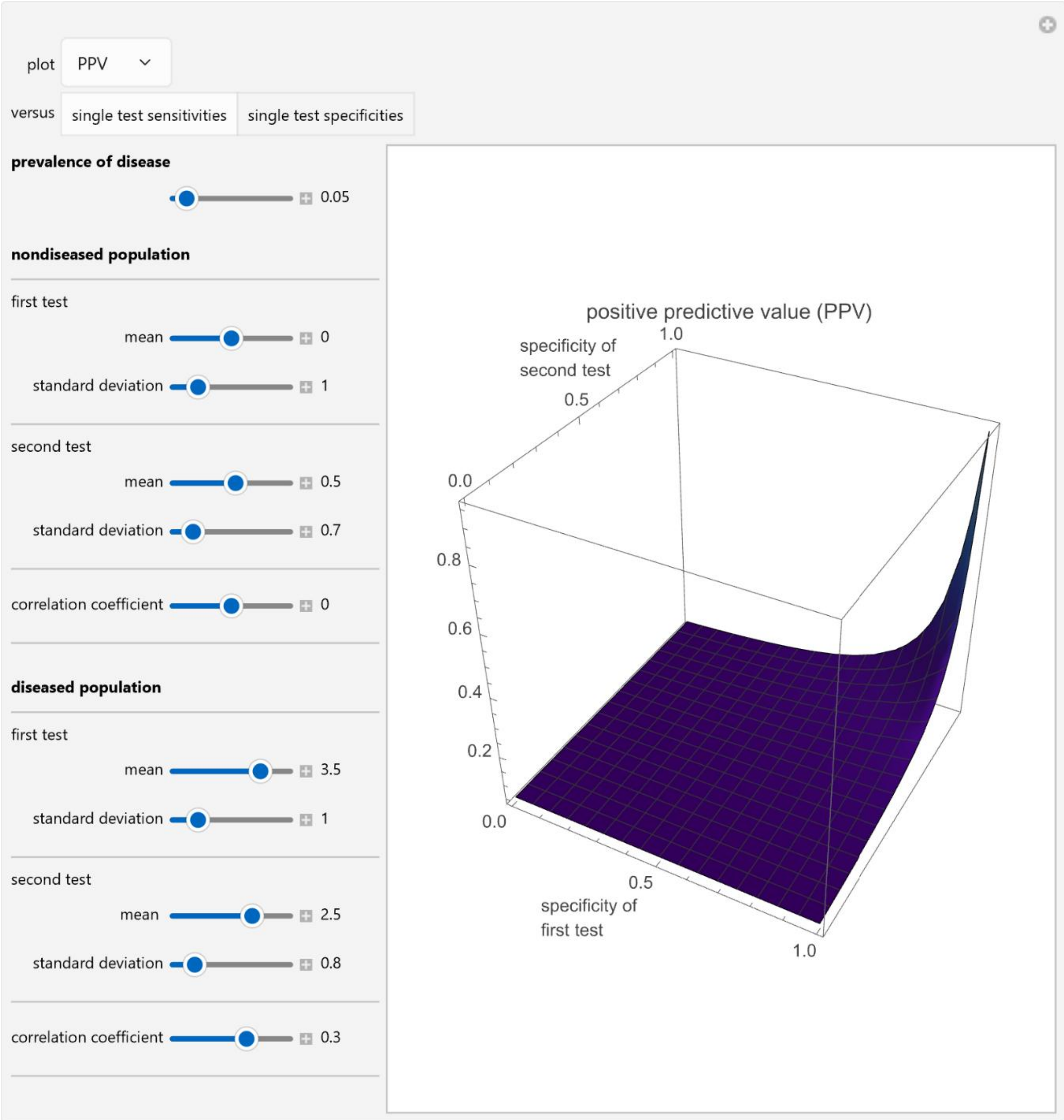

Figure 3: *Plot of the positive predictive value of two combined diagnostic tests plotted against the specificity of each single test, with the settings shown on the left.*

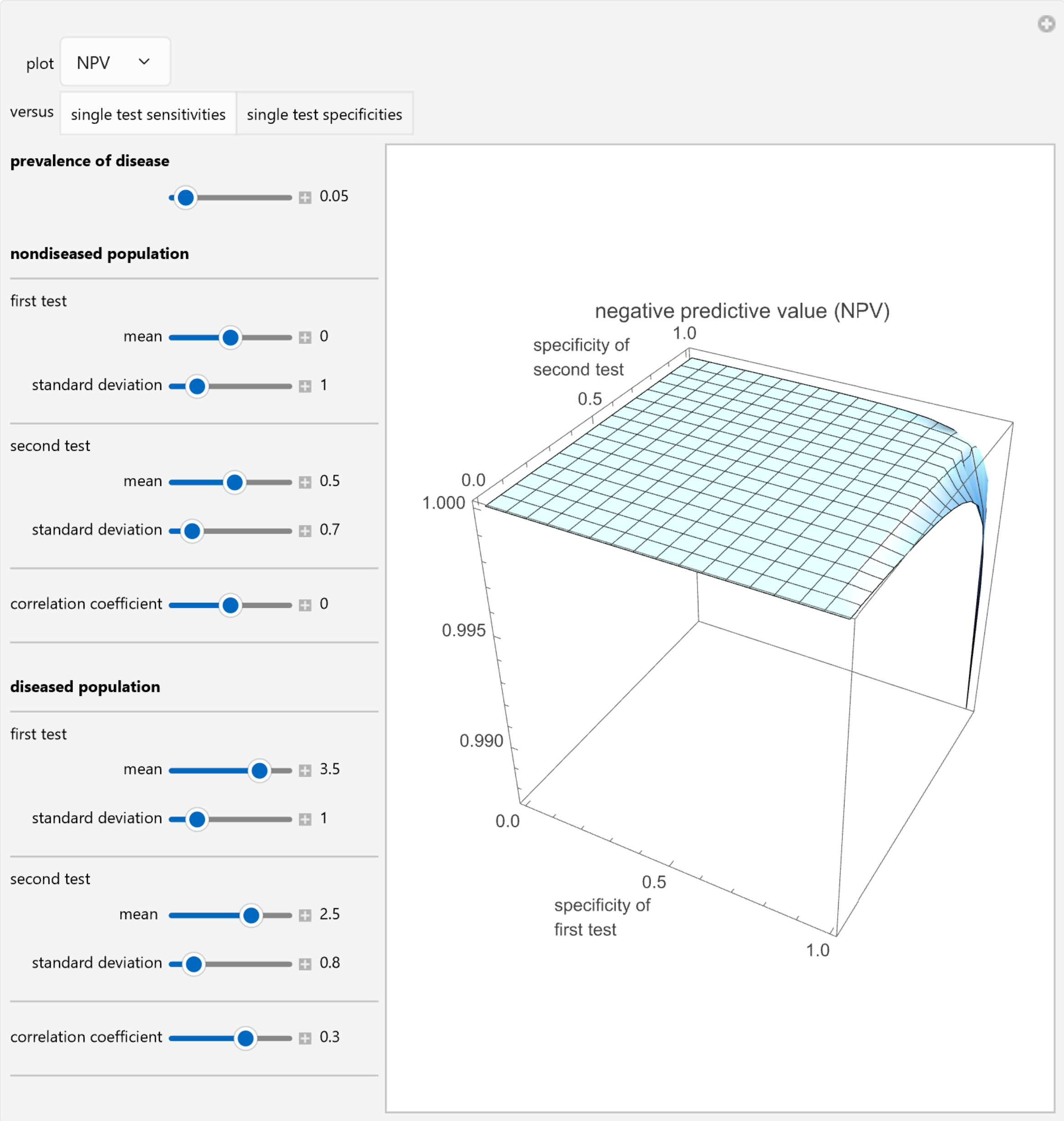

Figure 4: *Plot of the negative predictive value of two combined diagnostic tests plotted against the specificity of each single test, with the settings shown on the left.*

Details

Assume bivariate normal distributions for the paired measurements for each of the two tests on both nondiseased and diseased populations. The measures are used in the evaluation of the diagnostic accuracy of the combined diagnostic tests applied to diseased or nondiseased populations. They can be calculated as functions of the sensitivities or the specificities of each of the two tests.

Sensitivity of each single test is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. Sensitivity of the two combined tests is the fraction of the diseased population with at least one positive test result, while specificity is the fraction of the nondiseased population with negative results for both tests. If we denote by CSe and CSp the sensitivity and the specificity of the combined tests, and by v the disease prevalence, we have:

$$PPV = \frac{CSe\ v}{CSe\ v + (1 - CSp)(1 - pr)}$$

$$NPV = \frac{CSp\ (1 - v)}{CSp\ (1 - v) + (1 - CSe)\ v}$$

$$OR = \frac{\frac{CSe}{1 - CSe}}{\frac{1 - CSp}{CSp}}$$

$$LR+ = \frac{CSe}{1 - CSp}$$

$$LR- = \frac{1 - CSe}{CSp}$$

Demonstration

Version

1.1.2

DOI

[10.5281/zenodo.20561046](https://doi.org/10.5281/zenodo.20561046)

Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/AnalysisOfDiagnosticAccuracyMeasuresForTwoCombinedDiagnostic.nb

First published on April 3, 2018, as part of the Wolfram Demonstrations Project.

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

Citation

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